
Casper

**Project Documentation:
Image-to-TikZ Engine**

Version 1.0

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3.2 TeX Live Compiler Adapter

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Bibliography

- [1] Robert C. Martin. *Clean Architecture: A Craftsman's Guide to Software Structure and Design*. Prentice Hall, 2017.
- [2] Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep Learning*. MIT Press, 2016.